Introduction to Linux Systems

Package Management

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Lab

- Install ROS
  - ROS Noetic for Ubuntu 20.04
  - **ROS2 Humble for Ubuntu 22.04**
  - ROS2 Jazzy for Ubuntu 24.04

Distribution: Humble → Ubuntu 22.04

Ubuntu (deb packages) — ROS 2 Documentation: Humble documentation

- Set locale

```
$ locale  # check for UTF-8

$ sudo apt update && sudo apt install locales
$ sudo locale-gen en_US en_US.UTF-8
$ sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
$ export LANG=en_US.UTF-8

$ locale  # verify settings
```

Distribution: Humble → Ubuntu 22.04

Ubuntu (deb packages) — ROS 2 Documentation: Humble documentation

- Setup Sources

```
$ sudo apt install software-properties-common
$ sudo add-apt-repository universe

$ sudo apt update && sudo apt install curl -y
$ export ROS_APT_SOURCE_VERSION=$(curl -s
https://api.github.com/repos/ros-infrastructure/ros-apt-source/releases/latest | grep -F
"tag_name" | awk -F\" '{print $4}')
$ curl -L -o /tmp/ros2-apt-source.deb
"https://github.com/ros-infrastructure/ros-apt-source/releases/download/${ROS_APT_SOURCE_VERSI
ON}/ros2-apt-source_${ROS_APT_SOURCE_VERSION}.$(. /etc/os-release && echo
${UBUNTU_CODENAME:-${VERSION_CODENAME}})_all.deb"
$ sudo dpkg -i /tmp/ros2-apt-source.deb
```

Distribution: Humble → Ubuntu 22.04

Ubuntu (deb packages) — ROS 2 Documentation: Humble documentation

- Install ROS 2 packages

```
$ sudo apt update
$ sudo apt upgrade
$ sudo apt install ros-humble-desktop
$ sudo apt install ros-dev-tools

$ source /opt/ros/humble/setup.bash
```

Demo - Try some examples

- First terminal

```
joe@Evan-Lab:~$ ros2 run demo_nodes_cpp talker
[INFO] [1762240291.103145463] [talker]: Publishing: 'Hello World: 1'
[INFO] [1762240292.071526577] [talker]: Publishing: 'Hello World: 2'
[INFO] [1762240293.041154373] [talker]: Publishing: 'Hello World: 3'
[INFO] [1762240294.007990111] [talker]: Publishing: 'Hello World: 4'
[INFO] [1762240294.972012314] [talker]: Publishing: 'Hello World: 5'
[INFO] [1762240295.935893378] [talker]: Publishing: 'Hello World: 6'
[INFO] [1762240296.900091956] [talker]: Publishing: 'Hello World: 7'
[INFO] [1762240297.864193756] [talker]: Publishing: 'Hello World: 8'
[INFO] [1762240298.828181174] [talker]: Publishing: 'Hello World: 9'
[INFO] [1762240299.792251354] [talker]: Publishing: 'Hello World: 10'
[INFO] [1762240300.756261149] [talker]: Publishing: 'Hello World: 11'
[INFO] [1762240301.712553013] [talker]: Publishing: 'Hello World: 12'
```

繳交作業需包含以下部分:

1. 畫面截圖
2. 本次作業/課程心得

將上述內容彙整成一份 PDF 檔案

格式:**HW7_*StudentID*.pdf**
e.g., **HW7_F74131234.pdf**

- Second terminal

```
joe@Evan-Lab:~$ ros2 run demo_nodes_py listener
[INFO] [1762240334.740561564] [listener]: I heard: [Hello World: 45]
[INFO] [1762240335.675550860] [listener]: I heard: [Hello World: 46]
[INFO] [1762240336.621824066] [listener]: I heard: [Hello World: 47]
[INFO] [1762240337.568340421] [listener]: I heard: [Hello World: 48]
[INFO] [1762240338.514862023] [listener]: I heard: [Hello World: 49]
[INFO] [1762240339.461761977] [listener]: I heard: [Hello World: 50]
[INFO] [1762240340.409711249] [listener]: I heard: [Hello World: 51]
[INFO] [1762240341.408518025] [listener]: I heard: [Hello World: 52]
```

QUESTIONS